Saad Dahlab University of Blida
Faculty of Science

May 16, 2026
1st Year LMD (C.S)

Last name:	First name:	Group:	Mark: / 20

Electronic Fundamentals Exam

Exercise 1 (6pts)

Consider the circuit presented in **Figure 1**. A load resistor R_4 is connected between the output terminals **A** and **B**.
Given: $E_1 = 4$ V, $R_1 = 16\,\Omega$, $R_2 = 4\,\Omega$, $R_3 = 6\,\Omega$, and $R_4 = 24\,\Omega$.

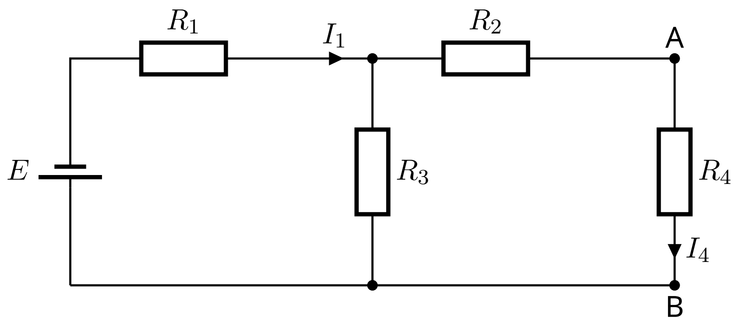


Figure 1

1. Determine the expression and calculate the equivalent resistance R_{eq} as a function of R_1, R_2, R_3 and R_4. ....

....

....

2. Calculate the current I_1 using Kirchhoff's Voltage Law (loop law). ....

....

3. Using Figure 1 and applying Norton's theorem, determine the expression of the Norton current I_N in terms of I_1, R_2, and R_3 as seen across the load resistor R_4. ....

....

....

4. Calculate the value of the Norton current I_N. ....

....

5. Give the expression of Norton resistance R_N and calculate its numerical value. ....

....

Exercise 2 (5pts)

- *Four options are provided as possible answers to the following questions. each question has only ONE correct answer. Choose the correct answer and write the letter (A-D) next to the question.*

Consider the circuit shown in **Figure 2**. The diodes D_1 and D_2 are ideal diodes. The input voltage is:

$$e(t) = 100\sqrt{2}\,\text{sin}(100\pi t)$$

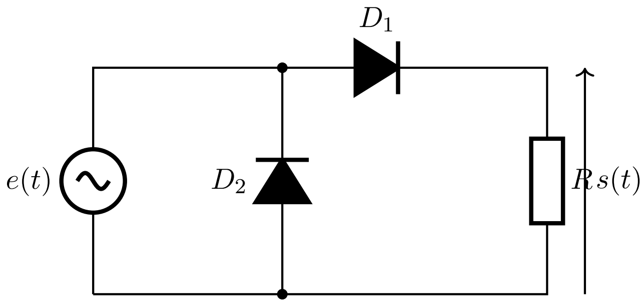


Figure 2

1. **When $e(t) > 0$, the diodes are:**

   A. D_1 conducting and D_2 blocked
   B. D_1 blocked and D_2 conducting
   C. Both diodes conducting
   D. Both diodes blocked

2. **When $e(t) < 0$, the diodes are:**

   A. D_1 conducting and D_2 blocked
   B. D_1 blocked and D_2 conducting
   C. Both diodes conducting
   D. Both diodes blocked

3. **The output voltage $s(t)$ is equal to:**

   A. $e(t)$ for all values of t
   B. 0 for all values of t
   C. $e(t)$ when $e(t) > 0$, and 0 when $e(t) < 0$
   D. $-e(t)$ when $e(t) > 0$

4. **For $t = \frac{1}{400}$ s, determine the value of the resistance R if the effective current through the resistor is equal to 10 mA.**

   A. 1 kΩ
   B. 5 kΩ
   C. 10 kΩ
   D. 51 kΩ

5. **The maximum current through R is:**

A. 14.14 mA

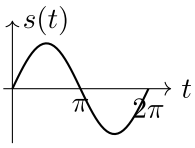
B. 11.11 mA

C. 20.20 mA

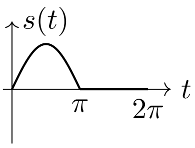
D. 19.19 mA

6. **The output signal $s(t)$ is shown below. Which graph correctly represents $s(t)$?**

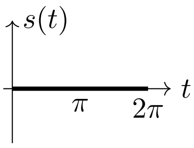
A.



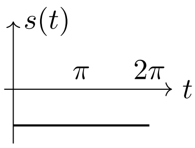
B.



C.

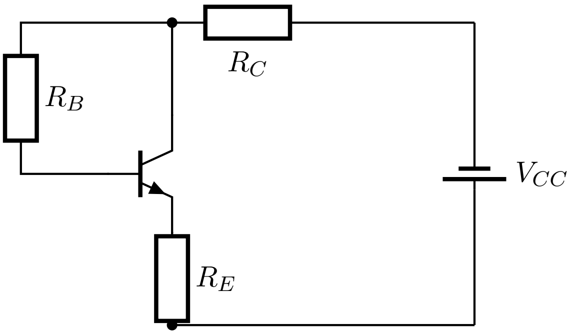


D.



Exercise 3 (9pts)

A silicon NPN transistor is biased using a resistor connected between the collector and the base, as shown in the diagram.



1. On the circuit diagram, identify the transistor terminals: base (B), collector (C), and emitter (E).

2. Indicate on the diagram the directions of the voltages V_{CE}, V_{CB}, and V_{BE}.

3. Show on the circuit diagram the directions of the currents I_B, I_C, and I_E.

4. Give the relationships between these three currents (I_B, I_C, and I_E). ....

   ....

5. Establish the equation of the DC load line $I_C = f(V_{CE})$ (output characteristic). ....

   ....

   ....

   ....

6. Determine the saturation current I_{Csat} corresponding to the condition $V_{CE} = 0$V. ....

....

7. Determine the cutoff voltage V_{CE} corresponding to the condition $I_C = 0$A. ....

....

8. Establish the equation of the input line $I_B = f(V_{BE})$ (input characteristic). ....

....

....

9. Plot the DC load line $I_C = f(V_{CE})$ (output characteristic). ....

....

....

....

....

....

10. Calculate the values of I_C and I_B required so that $V_{CE} = 5$V, while keeping the other parameters unchanged. ....

....

....

....

Given:
$\beta = 65$, $V_{CC} = 10$V, $V_{BE} = 0.7$V, $R_B = 17$ kΩ, $R_C = 1$ kΩ, and $R_E = 100\,\Omega$.